Combinatorics: 20 Worked Problems

Fundamental Counting · Permutations & Circular Arrangements · Stars & Bars · Inclusion–Exclusion & Derangements · Catalan Numbers, Compositions & Partitions

These 20 original problems are built to drill exactly the sub-skills listed under *Notebook 1, Chapter 1.1* of your syllabus. Every solution is a full derivation — no hand-waving — and every problem carries a boxed *Speed Trick* so you can also solve it the Stage-1 way, in under 90 seconds. Work Section A → E in order: each section leans on the identities proved in the one before it.

A. Fundamental Counting Rules (Sum & Product)

Problem A1 (Product rule with restricted positions). A trading-floor badge code is 2 letters from $\{A, \dots, Z\}$ followed by 3 digits from $\{0, \dots, 9\}$. The first letter cannot be Q, and the last digit cannot be 0. How many valid codes exist?

Solution. Positions are filled independently, so the product rule applies directly:

$$\underbrace{25}_{\text{1st letter}} \times \underbrace{26}_{\text{2nd letter}} \times \underbrace{10}_{\text{1st digit}} \times \underbrace{10}_{\text{2nd digit}} \times \underbrace{9}_{\text{3rd digit}} = 585{,}000.$$

▶ *Speed trick (Stage 1, <90s):* Restricted slots multiply in exactly where they sit; never subtract restrictions at the end unless the restriction couples two positions together.

Problem A2 (Sum rule over disjoint cases, each built by the product rule). A trader's lunch is either (i) a wrap (5 types) paired with a side (3 types), or (ii) a salad (4 types, no side). How many distinct lunch orders are possible?

Solution. The two cases are mutually exclusive (a wrap-order and a salad-order can never coincide), so by the rule of sum we simply add the two product-rule counts:

$$\underbrace{5 \times 3}_{\text{wrap branch}} + \underbrace{4}_{\text{salad branch}} = 15 + 4 = 19.$$

▶ *Speed trick (Stage 1, <90s):* Sum rule = "OR" between *disjoint* branches; product rule = "AND" *within* a branch. Mixed problems almost always decompose into product-rule branches added by the sum rule.

Problem A3 (Nested branch counting). A terminal PIN starts by picking an icon: 4 "Land" icons or 2 "Sea" icons. A Land icon is followed by a 4-digit numeric code (digits 0–9, repeats allowed); a Sea icon is followed by a 3-letter code (repeats allowed). How many total PINs exist?

Solution. Branches are disjoint (an icon is Land *or* Sea, never both), and within each branch the icon choice and the code choice are independent:

$$\underbrace{4 \times 10^4}_{\text{Land}=40{,}000} + \underbrace{2 \times 26^3}_{\text{Sea}=35{,}152} = 75{,}152.$$

▶ *Speed trick (Stage 1, <90s):* Draw the two-level tree mentally: multiply along a branch, add across branches. This "tree = product along, sum across" heuristic solves 90% of Stage-1 counting questions instantly.

B. Permutations, Combinations & Circular Arrangements

Problem B1 (Circular permutation with an adjacency restriction). Eight researchers, including twins Ravi and Rohan, sit around a circular table (rotations counted as identical). In how many seatings are Ravi and Rohan *not* adjacent?

Solution. Circular arrangements of n distinct people $= (n-1)!$, so all seatings $= (8-1)! = 5040$. Glue Ravi–Rohan into one block to count the *adjacent* seatings: this leaves 7 units around the table, giving $(7-1)! = 720$ circular arrangements, times 2! for the two internal orders of the block:

$$720 \times 2 = 1440 \text{ (adjacent seatings).}$$

By complementation:

$$5040 - 1440 = \boxed{3600}.$$

▶ *Speed trick (Stage 1, <90s):* "Not adjacent" in a circular table ⇒ always compute (total) − (block them together). Never try to place the gap directly; it is far more error-prone than complementation.

Problem B2 (Pascal's identity, proved combinatorially). Prove $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ by a counting argument, then use it to double-check a subcommittee count: from a 12-person desk that includes a Head Trader H, how many 5-person subcommittees are there?

Solution. *Proof.* $\binom{n}{k}$ counts k-subsets of an n-set. Fix one element x. Every k-subset either (i) contains x — choose the remaining $k-1$ members from the other $n-1$ elements, $\binom{n-1}{k-1}$ ways — or (ii) excludes x — choose all k members from the other $n-1$ elements, $\binom{n-1}{k}$ ways. These cases are disjoint and exhaustive, so by the sum rule $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$. ■

Check. Direct count: $\binom{12}{5} = 792$. Split on whether H is included: $\binom{11}{4} + \binom{11}{5} = 330 + 462 = 792$. ✓

▶ *Speed trick (Stage 1, <90s):* Whenever a problem says "how many committees include a specific person / exclude a specific person," you are being handed Pascal's identity as a shortcut — split, don't enumerate.

Problem B3 (Binomial coefficients as lattice-path counts). A trade-execution path moves only Right (R) or Up (U) on a grid. How many paths run from $(0,0)$ to $(5,3)$? How many of those pass through the checkpoint $(2,1)$?

Solution. A path to $(5,3)$ is a string of 5 R's and 3 U's in some order; choosing *which* 3 of the 8 steps are U's determines the whole path:

$$\binom{8}{3} = 56 \text{ total paths.}$$

A path through $(2,1)$ is a path to $(2,1)$ (needs $\binom{3}{1} = 3$ arrangements) followed independently by a path from $(2,1)$ to $(5,3)$, i.e. to $(3,2)$ relative to the checkpoint ($\binom{5}{2} = 10$ arrangements). By the product rule:

$$3 \times 10 = 30.$$

▶ *Speed trick (Stage 1, <90s):* Grid-path counting is just "choose the positions of the U's." Checkpoints split the path into two independent sub-problems multiplied together — this is the exact combinatorial seed of the reflection principle you'll use for Catalan numbers in Section E.

Problem B4 (Non-adjacent selection — the "gap" bijection). How many ways can you choose 4 non-consecutive timeslots from 15 slots numbered $1, \dots, 15$ (no two chosen slots may be adjacent)?

Solution. Let the chosen slots in increasing order be $x_1 < x_2 < x_3 < x_4$ with $x_{i+1} - x_i \geq 2$. Define $y_i = x_i - (i-1)$. Then $y_1 < y_2 < y_3 < y_4$ with $y_{i+1} - y_i \geq 1$ (an ordinary strictly increasing sequence), and y_i ranges over $\{1, \dots, 15-3\} = \{1, \dots, 12\}$. This is a bijection onto ordinary 4-subsets of a 12-element set:

$$\binom{15-4+1}{4} = \binom{12}{4} = 495.$$

▶ *Speed trick (Stage 1, <90s):* General formula: choosing k non-adjacent items from n in a row = $\binom{n-k+1}{k}$. (On a *circle* of n instead of a row, the analogous count is $\frac{n}{n-k}\binom{n-k}{k}$ — memorize both for Stage 1.)

C. Stars and Bars (Balls into Urns)

Problem C1 (Basic weak composition, $x_i \geq 0$). In how many ways can 10 identical lot-tickets be distributed among 4 trading desks (a desk may receive zero)?

Solution. Number of non-negative integer solutions to $x_1 + x_2 + x_3 + x_4 = 10$ is the classic stars-and-bars count: arrange 10 stars and $4 - 1 = 3$ bars in a row:

$$\binom{10+4-1}{4-1} = \binom{13}{3} = 286.$$

▶ *Speed trick (Stage 1, <90s):* $\binom{n+k-1}{k-1}$ — always n stars, $k-1$ bars, and the answer is "choose where the bars go" among $n+k-1$ total symbols.

Problem C2 (Lower-bound constraint, $x_i \geq 1$). Same as C1, but every desk must receive at least one ticket.

Solution. Substitute $y_i = x_i - 1 \geq 0$. Then $\sum y_i = 10 - 4 = 6$, distributed among 4 desks:

$$\binom{6+4-1}{4-1} = \binom{9}{3} = 84.$$

▶ *Speed trick (Stage 1, <90s):* "At least 1 each" always means: subtract 1 from n per bin *before* applying the formula, i.e. use $\binom{(n-k)+k-1}{k-1} = \binom{n-1}{k-1}$ directly. Sanity check: $\binom{10-1}{4-1} = \binom{9}{3} = 84$. ✓

Problem C3 (Capacity cap via inclusion–exclusion). Distribute 12 identical risk-units among 3 desks so that *no desk receives more than 6*. How many allocations?

Solution. Unrestricted count: $\binom{12+3-1}{3-1} = \binom{14}{2} = 91$. Let B_i = "desk i gets ≥ 7." Substituting $x_i' = x_i - 7 \geq 0$ turns $\sum x_j = 12$ into $\sum x_j' = 5$ (3 vars), giving $\binom{5+2}{2} = 21$ solutions in each B_i, and there are 3 desks, so $\sum |B_i| = 3 \times 21 = 63$. Two desks cannot both exceed 6 simultaneously since $7+7 = 14 > 12$, so all pairwise (and triple) intersections are empty — no correction terms needed. By inclusion–exclusion:

$$91 - 63 = \boxed{28}.$$

▶ *Speed trick (Stage 1, <90s):* Capacity caps are stars-and-bars *minus* PIE on the "overflow" event $x_i \geq \text{cap} + 1$. Always check whether two overflow events can coexist (compare $2(\text{cap}+1)$ to the total) before assuming higher PIE terms vanish.

Problem C4 (Parity-constrained composition). In how many ways can 20 be written as an ordered sum of 5 non-negative *even* integers?

Solution. Write $x_i = 2y_i$ with $y_i \geq 0$. Then $2\sum y_i = 20 \Rightarrow \sum y_i = 10$ over 5 variables:

$$\binom{10+5-1}{5-1} = \binom{14}{4} = 1001.$$

▶ *Speed trick (Stage 1, <90s):* A "multiples of m" constraint always factors out: substitute $x_i = my_i$, divide the target by m, and re-apply the ordinary stars-and-bars formula to $y_i \geq 0$.

Problem C5 (Stars and bars inside a probability computation). $n = 10$ identical buy-signals are routed to $k = 4$ desks, with *every* weak composition equally likely. Find $P(\text{desk 1 receives exactly } j = 3 \text{ signals})$.

Solution. Sample space size (all weak compositions of 10 into 4 parts): $\binom{10+4-1}{4-1} = \binom{13}{3} = 286$. Favorable outcomes: fix $x_1 = 3$, then distribute the remaining 7 among the other 3 desks: $\binom{7+3-1}{3-1} = \binom{9}{2} = 36$.

$$P = \frac{36}{286} = \frac{18}{143} \approx 0.1259.$$

▶ *Speed trick (Stage 1, <90s):* "Fix one bin's value, re-run stars-and-bars on the rest" is the standard way to get a marginal distribution out of a uniform-composition model. Note this is *not* the Binomial distribution — that would apply only if each signal independently and uniformly chose a desk.

D. Inclusion–Exclusion & Derangements

Problem D1 (Two-set PIE, mental-math speed). Among the integers 1 to 900, how many are divisible by neither 15 nor 24?

Solution. $\lfloor 900/15 \rfloor = 60$, $\lfloor 900/24 \rfloor = 37$, $\text{lcm}(15, 24) = 120 \Rightarrow \lfloor 900/120 \rfloor = 7$.

$$|A \cup B| = 60 + 37 - 7 = 90 \;\Rightarrow\; 900 - 90 = \boxed{810}.$$

▶ *Speed trick (Stage 1, <90s):* Always convert "divisible by a and b" into "divisible by $\text{lcm}(a, b)$" — never $a \times b$ unless $\gcd(a, b) = 1$. This is the single most common Stage-1 PIE error.

Problem D2 (Three-set PIE). Of 200 traders surveyed: 120 trade equities (E), 90 trade FX (F), 70 trade commodities (C); $|E \cap F| = 50$, $|F \cap C| = 40$, $|E \cap C| = 30$; $|E \cap F \cap C| = 15$. How many trade none of the three?

Solution.

$$|E \cup F \cup C| = 120 + 90 + 70 - 50 - 40 - 30 + 15 = 175.$$

$$\text{None} = 200 - 175 = \boxed{25}.$$

▶ *Speed trick (Stage 1, <90s):* Memorize the PIE sign pattern as "+singles −pairs +triples −⋯" and always list the six numbers in that exact order before adding — misplacing one sign is the most common error under time pressure.

Problem D3 (Full derangement derivation). Derive the derangement formula $D_n = n! \sum_{i=0}^{n} \frac{(-1)^i}{i!}$ via PIE, then compute D_7 and the probability that none of 7 analysts receives their own correctly labeled report.

Solution. Let $A_i \subset S_n$ be the set of permutations fixing position i. We want $n! - |A_1 \cup \cdots \cup A_n|$. For any r indices, $|A_{i_1} \cap \cdots \cap A_{i_r}| = (n-r)!$ (the other $n-r$ positions permute freely), and there are $\binom{n}{r}$ such intersections. By PIE:

$$|A_1 \cup \cdots \cup A_n| = \sum_{r=1}^{n} (-1)^{r+1} \binom{n}{r} (n-r)!$$

$$D_n = n! - \sum_{r=1}^{n} (-1)^{r+1} \binom{n}{r} (n-r)! = \sum_{r=0}^{n} (-1)^r \binom{n}{r} (n-r)!$$

Since $\binom{n}{r}(n-r)! = \frac{n!}{r!}$, this collapses to

$$D_n = n! \sum_{r=0}^{n} \frac{(-1)^r}{r!}. \qquad \blacksquare$$

For $n = 7$: $D_7 = 5040\left(1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} + \frac{1}{720} - \frac{1}{5040}\right) = 1854$.

$$P(\text{no matches}) = \frac{1854}{5040} \approx 0.3679 \approx \frac{1}{e}.$$

▶ *Speed trick (Stage 1, <90s):* For $n \geq 7$ or so, skip the sum entirely: $D_n \approx n!/e$, rounded to the nearest integer. It's exact for all practical Stage-1 values (error $< 1/(n+1)$).

Problem D4 (Bonferroni bound — fast mental estimate). Five roughly independent trading alerts each fire with probability 0.02. Bound $P(\text{at least one fires})$ using Boole's inequality, then compare with the exact independent-event value.

Solution. *Boole (first-order) upper bound:*

$$P\left(\bigcup A_i\right) \le \sum p_i = 5(0.02) = 0.10.$$

Exact value (independent events):

$$P\left(\bigcup A_i\right) = 1 - (0.98)^5 = 1 - 0.9039 = 0.0961.$$

The first-order bound overshoots by only $\approx 4\%$ relative error here because the p_i are small, so pairwise-overlap "double counting" is second-order small $\binom{5}{2}(0.02)^2 = 0.004$, matching the gap almost exactly).

▶ *Speed trick (Stage 1, <90s):* When all $p_i \le 0.05$ or so, Boole's bound $\sum p_i$ *is* your Stage-1 answer to 2 significant figures — don't waste time on the exact complement unless the question explicitly demands precision.

E. Catalan Numbers, Compositions & Partitions

Problem E1 (Catalan number via the reflection principle — Dyck paths). A market-maker executes 8 trades: 4 buys (+1) and 4 sells (−1), in some order. In how many orders does the running inventory never go negative?

Solution. Total orderings of 4 buys and 4 sells: $\binom{8}{4} = 70$. *Reflection principle.* A "bad" path touches -1 at some point. Reflect the portion of the path up to its first visit to -1 across the line $y = -1$: this maps every bad path bijectively onto *every* path of 4 buys/4 sells that ends at height -2 instead of 0, i.e. a path with 3 buys and 5 sells: $\binom{8}{3} = 56$.

$$\text{Good paths} = 70 - 56 = 14.$$

This matches the closed form $C_4 = \dfrac{1}{4+1}\binom{8}{4} = \dfrac{70}{5} = 14.$

▶ *Speed trick (Stage 1, <90s):* $C_n = \dfrac{1}{n+1}\binom{2n}{n}$ is "total arrangements $\div (n+1)$" — for balance/ranking/non-negativity problems with n up-steps and n down-steps, this is almost always the final answer.

Problem E2 (Catalan number — balanced matching). How many ways can 5 pairs of order confirmations (a "sell" must match a strictly earlier, still-unmatched "buy" — no crossing allowed) be validly matched, i.e. how many balanced bracket sequences of length 10 are there?

Solution. This is exactly the Dyck-path count for $n = 5$:

$$C_5 = \frac{1}{6}\binom{10}{5} = \frac{252}{6} = 42.$$

▶ *Speed trick (Stage 1, <90s):* Recognize the phrase "every closing element matches a strictly earlier, not-yet-matched opening element" — that sentence, in any costume (parentheses, buy/sell confirmations, Dyck paths, binary tree shapes), always means "compute C_n."

Problem E3 (Compositions — ordered positive parts). In how many ways can a trader split a \$50,000 position into 4 sequential, strictly positive tickets sized in whole \$1000 units?

Solution. We need ordered positive integer solutions (a *composition*) to $t_1 + t_2 + t_3 + t_4 = 50$, $t_i \ge 1$. Place 50 stars in a row; a composition into 4 positive parts corresponds to choosing 3 of the 49 internal gaps to hold a bar:

$$\binom{50-1}{4-1} = \binom{49}{3} = 18{,}424.$$

▶ *Speed trick (Stage 1, <90s):* Compositions of n into k positive parts $= \binom{n-1}{k-1}$ — it's the $k-1$ *internal gaps* of n stars, not the $n+k-1$ symbols you use for the $x_i \ge 0$ version. Confusing these two binomial coefficients is the #1 stars-and-bars trap.

Problem E4 (Partitions vs. compositions — the classic dice-sum trap). In how many ways can 6 be written as a sum of *at most 3* positive integers where order does **not** matter (a partition)? Contrast this with the number of *ordered* (composition) solutions using exactly 3 positive parts.

Solution. *Partitions of 6 into at most 3 parts* (direct enumeration — no closed form exists for partitions, unlike compositions):

$$6;\quad 5{+}1;\quad 4{+}2;\quad 3{+}3;\quad 4{+}1{+}1;\quad 3{+}2{+}1;\quad 2{+}2{+}2.$$

That's $\boxed{7}$ partitions. *Compositions of 6 into exactly 3 positive parts* (order matters): $\binom{6-1}{3-1} = \binom{5}{2} = 10$ — e.g. 3+2+1, 1+2+3, 2+1+3, … are counted as 6 *different* compositions but only 1 partition.

▶ *Speed trick (Stage 1, <90s):* "Three indistinguishable dice summing to s" is a **partition** question (enumerate by hand for small s — there's no formula). "Three distinguishable/ordered dice summing to s" is a **composition/stars-and-bars** question and has the closed form above. Reading which one the problem means, correctly and instantly, is worth more Stage-1 points than any formula.

Drill order for review: re-derive B2, C3, D3, and E1 from scratch (closed-book) — these four carry the four reusable tricks (Pascal split, capacity-cap PIE, full derangement PIE, reflection principle) that generalize to almost every Stage-1 combinatorics question outside this sheet.